Concavity, Curve Sketching, and Optimization

ANSWER KEY

Section 1: Concavity & Points of Inflection

1 On which interval is the function $f(x) = x^3 - 6x^2 + 9x$ concave up?

a. $(2, \infty)$ ✓

b. $(-\infty, 2)$

c. $(1, 3)$

d. $(0, \infty)$

$f'(x) = 3x^2 - 12x + 9$. $f''(x) = 6x - 12$. Concave up where $f''(x) > 0 \Rightarrow 6x - 12 > 0 \Rightarrow x > 2$.

2 What is the point of inflection of the curve $f(x) = x^3 - 6x^2 + 9x$?

a. $(2, 2)$ ✓

b. $(2, 0)$

c. $(1, 4)$

d. $(3, 0)$

$f''(x) = 0 \Rightarrow x = 2$. $f(2) = 2^3 - 6(4) + 9(2) = 8 - 24 + 18 = 2$. Since f'' changes sign across $x = 2$, $(2, 2)$ is a point of inflection.

3 What is a necessary condition for $(c, f(c))$ to be a point of inflection?

a. $f''(c) = 0$ **or** $f''(c)$ **undefined, AND** f'' **changes sign across** c ✓

b. $f'(c) = 0$

c. $f''(c) = 0$ only

d. $f(c) = 0$

A point of inflection requires the concavity to change sign, meaning f'' must change sign across c (where $f''(c) = 0$ or does not exist).

Section 2: Second Derivative Test for Local Extrema

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4 If $f'(3) = 0$ and $f''(3) = -4$, what does the Second Derivative Test conclude about $x = 3$?

a. f **has a local maximum at** $x = 3$ ✓

b. f has a local minimum at $x = 3$

c. The test is inconclusive

d. $x = 3$ is an inflection point

Since $f'(3) = 0$ (horizontal tangent) and $f''(3) < 0$ (concave down), f has a local maximum at $x = 3$.

5 When is the Second Derivative Test inconclusive at a critical point c?

a. **When** $f''(c) = 0$ **or** $f''(c)$ **does not exist** ✓

b. When $f''(c) > 0$

c. When $f'(c) = 0$

d. When $f(c) = 0$

If $f''(c) = 0$ or fails to exist, the test provides no information and one must use the First Derivative Test instead.

6 Apply the Second Derivative Test to $f(x) = x^4 - 4x^3$ at its critical point $x = 3$.

a. **Local minimum because** $f''(3) = 36 > 0$ ✓

b. Local maximum because $f''(3) < 0$

c. Inflection point

d. Inconclusive

$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3) = 0 \Rightarrow x = 3.$
$f''(x) = 12x^2 - 24x \Rightarrow f''(3) = 12(9) - 24(3) = 108 - 72 = 36 > 0.$ Local minimum.

Section 3: Applied Optimization: Geometric Problems

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7 A farmer has 200 meters of fencing to enclose a rectangular field adjacent to a straight river (no fence needed along the river). What dimensions maximize the area?

a. **Length 100 m, Width 50 m** ✓

b. Length 80 m, Width 60 m

c. Length 120 m, Width 40 m

d. Square of side 66.7 m

Fence: $2W + L = 200 \Rightarrow L = 200 - 2W$. Area $A(W) = W(200 - 2W) = 200W - 2W^2$.
$A'(W) = 200 - 4W = 0 \Rightarrow W = 50$ m, $L = 100$ m.

8 An open-top rectangular box is formed by cutting equal squares of side x from the corners of a 12 by 12 cm sheet of cardboard. What value of x maximizes the box volume?

a. $x = 2$ **cm** ✓

b. $x = 3$ cm

c. $x = 1.5$ cm

d. $x = 4$ cm

$V(x) = x(12 - 2x)^2 = x(144 - 48x + 4x^2) = 4x^3 - 48x^2 + 144x$.
$V'(x) = 12x^2 - 96x + 144 = 12(x^2 - 8x + 12) = 12(x - 2)(x - 6) = 0$. Since $0 < x < 6$, $x = 2$ cm.

9 What is the maximum volume in the box problem above?

a. 128 **cm**^{3} ✓

b. 144 cm^3

c. 108 cm^3

d. 96 cm^3

$V(2) = 2(12 - 4)^2 = 2(8^2) = 2(64) = 128$ cm^3.

Section 4: Applied Optimization: Cost & Distance

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10 Which point on the line $y = 2x + 1$ is closest to the origin $(0, 0)$?

a. $(-\frac{2}{5}, \frac{1}{5})$ ✓

b. $(-1, -1)$

c. $(0, 1)$

d. $(-\frac{1}{2}, 0)$

Squared distance $D(x) = x^2 + (2x + 1)^2 = 5x^2 + 4x + 1$. $D'(x) = 10x + 4 = 0 \Rightarrow x = -\frac{2}{5}$. Then $y = 2(-2/5) + 1 = \frac{1}{5}$.

11 A company determines that total cost for producing x units is $C(x) = 0.01x^2 + 4x + 100$. What production level minimizes average cost $\bar{C}(x) = \frac{C(x)}{x}$?

a. $x = 100$ ✓

b. $x = 50$

c. $x = 200$

d. $x = 400$

$\bar{C}(x) = 0.01x + 4 + \frac{100}{x}$. $\bar{C}'(x) = 0.01 - \frac{100}{x^2} = 0 \Rightarrow x^2 = \frac{100}{0.01} = 10,000 \Rightarrow x = 100$.

12 Two positive numbers have a product of 36. What is their minimum sum?

a. 12 ✓

b. 13

c. 15

d. 20

Let numbers be x and $\frac{36}{x}$. $S(x) = x + \frac{36}{x}$. $S'(x) = 1 - \frac{36}{x^2} = 0 \Rightarrow x = 6$. Minimum sum $= 6 + 6 = 12$.

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